Masculinity Norms, Occupational Choice, and Reallocation: Micro and Macro Evidence

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Abstract

Over the past five decades, the labor force participation rate of American men has declined significantly—from 80% in the 1970s to just around 68% today—and similar trends are observed across the developed world. During the same period, the fastest-growing occupations have been those traditionally considered "feminine," such as nursing. Are these two trends connected? Do masculinity norms discourage men from entering the labor force by deterring them from pursuing jobs perceived as feminine? We provide evidence at the macro, micro, and experimental level. At the macro level, we document that male employment participation declined more following sectoral reallocation in more gender-traditional labor markets. At the micro level, we show that gender-progressive men's employment recovers somewhat faster after layoff than that of gender-traditional men, partly through switching into female-dominated industries and occupations. Finally, we are piloting a survey experiment designed to measure men's willingness to pay to avoid occupations perceived as feminine and to assess whether correcting misperceptions about masculinity norms changes this willingness to pay. Understanding the role of masculinity norms in shaping men's labor market decisions is crucial for addressing the ongoing decline in male labor force participation.

1 Introduction

Despite strong economic growth for most of the past 50 years, male labor force participation in the US has been steadily declining. At the same time, the majority of recent growth has been in female-dominated industries (Figure 1). We hypothesize that these two trends—declining male participation and female-biased employment growth—are related: if a man is unwilling or unable to work in a female-dominated industry, this growth, rather than being an opportunity, will drive him out of the market. The possibility that masculinity norms are strong enough that some men would rather exit the labor force entirely than enter a female-dominated occupation makes this question particularly urgent.

Our work is based on a three-pronged approach. At the *aggregate* level, we show that a reallocation of labor demand across industries has led to a sharper fall in male employment in areas with greater

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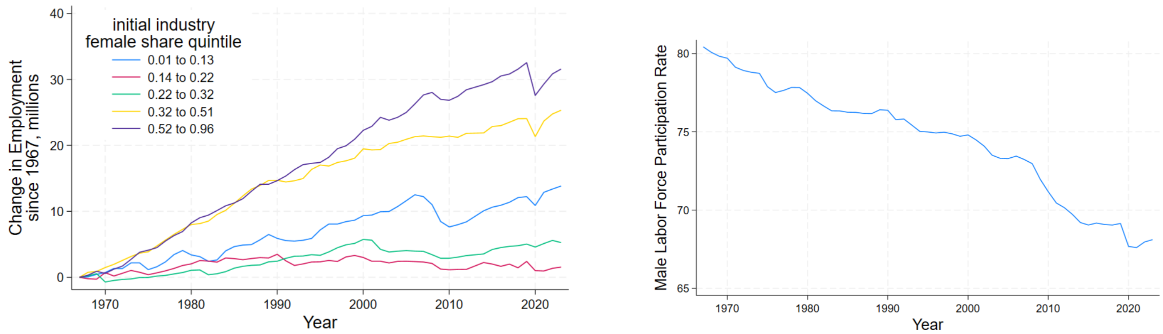


(a) Employment Growth by Female Share

(b) Male Labor Force Participation

Figure 1: Trends in the US Labor Market, 1967-2023

Note: The left panel shows the employment change of industries by their quintile of female share in 1968-1972 from the 1970s to today in the US (*Source: CPS Annual Social and Economic Supplement via IPUMS.*). The right panel shows male labor force participation over the same period (*Source: U.S. Bureau of Labor Statistics via FRED.*).

occupational segregation by gender, which are also the more gender-conservative. At the *individual* level, we examine how individual gender norms correlate with post-layoff employment recovery and with the occupational choices of the next generation. Finally, at the *experimental* level, we are piloting a survey experiment designed to measure the willingness to pay to avoid gender-atypical occupations and to test whether correcting misperceptions about masculinity norms can shift occupational preferences.

In all three cases, we find patterns consistent with our hypothesis that masculinity norms, by constraining occupation choice, act as a friction in male (re)employment.

At the aggregate level, we find that commuting zones that experienced a stronger sectoral reallocation towards more female jobs also experienced a larger decline in the employment-to-population ratio of men, but more so in more gender conservative areas. This is true in the raw data, where the correlation between growth of female-dominated sector and decline in male employment-to-population is twice as strong in conservative states than in progressive states. This also remains true once we include geographic and educational controls, instrument sectoral reallocation via a Bartik instrument, and replace state-level measures of gender norms with a commuting zone-level measure of gender industry segregation.

At the individual level, we find suggestive evidence that gender norms impact the speed at which men find a job after a layoff, as well as influencing one's children's occupation choice. Using data from the Panel Study of Income Dynamics (PSID), we find that men who have children with more gender-progressive women find re-employment somewhat faster after a layoff, through a greater probability of transitioning to a female occupation and to a female industry. We also find that children of mothers who agree with the statement *"There is some work that is men's and some that is women's and they should not be doing each other's"* exhibit greater occupational and industry segregation as adults, suggesting that norms are intergenerationally persistent and influence occupation choice.

The observational evidence is suggestive, but presents two issues: we only measure proxies of masculinity norms, and these proxies are presumably correlated with other factors that might confound our estimates; this is why we turn to experimental evidence. In a survey we are currently piloting we pursue three objectives: measuring (and experimentally varying) masculinity norms; computing willingness-to-pay to avoid gender-nonconforming occupations; and separating occupation preferences from existing human capital by asking about desired occupations for one's child.

The pilot evidence suggests the presence of pluralistic ignorance and a sizeable correlation between masculinity norms and willingness-to-pay to avoid feminine occupations. While 24% of respondents agree that *"Men should be aggressive and competitive to get ahead,"* the average estimate for the fraction of others who would agree is above 50%; and the bias is similarly large for all other five masculinity statements we elicit. In addition, men who agree with at least one of the six masculinity statements have a willingness-to-pay to avoid feminine occupations about 60% larger than men who agree with none. At this pilot stage, the difference is not statistically significant, but direction and magnitude point to a strong correlation between masculinity norms and occupation choice, and the presence of pluralistic ignorance suggests scope for a debiasing intervention.

Contribution. Our paper contributes to two strands of the literature. First, we contribute to the literature on worsening male labor market outcomes (Abraham and Kearney, 2020; Binder and Bound, 2019; Aguiar et al., 2021; Autor et al., 2019) and on structural transformation and reallocation that has favored women (Ngai and Petrongolo, 2017; Cortes et al., 2023). We provide the missing link between these two trends by showing that masculinity norms are a friction that prevents men from taking advantage of growth in female-dominated sectors. Second, we contribute to the growing literature on gender norms and labor market behavior, including work on masculinity norms (Matavelli et al., 2026; Baranov et al., 2023), gender norms more broadly (Cortés et al., 2026), and experimental evidence on social norms and labor supply (Bursztyn et al., 2020; Delfino, 2024). We add to this literature by linking experimental evidence with aggregate outcomes and by providing a numerical estimate of the willingness to pay to conform to gender norms in occupational choice.

2 Data

2.1 Census Data for Macro Analysis

We use data from the US Census to measure gender segregation and reallocation. Our samples cover 1980 through 2010, using the decennial censuses for 1980, 1990, and 2000, and pooling the 2009-2011 American Community Surveys for 2010. We define local areas using 1990 commuting zones. We define industries as broad groups of time-consistent 1950 industry codes[1], which gives us 20 total industries.

Reallocation We define reallocation in a local area a following Chodorow-Reich and Wieland (2020) (which in turn follows Lilien, 1982) as a measure of dispersion across industry employment growth rates. Reallocation in area a between period t and $t+j$ is then

$$R_{a,t,t+j} = \frac{1}{j}\frac{1}{2}\sum_{i}^{I} s_{a,i,t}\left|\frac{1+g_{a,i,t,t+j}}{1+g_{a,t,t+j}} - 1\right|, \tag{1}$$

where i indexes industries, $s_{a,i,t}$ is the employment share of industry i in area a at time t, and $g_{a,i,t,t+j}$ is the growth rate of industry i in area a between t and $t+j$. If all industries in an area grow at an equal pace, the main term of the index (before annualization) is equal to zero, and it is equal to one if all industries with positive employment in t disappear by $t+j$. Chodorow-Reich and Wieland (2020) find that reallocation contributes to higher local area unemployment if it occurs during a recession, but little difference in outcomes if it occurs during an expansion.

[1] *ind1950* in IPUMS

Growth of Female-Dominated Sectors We isolate growth in female-dominated sectors by computing the reallocation measure above, but only for "female" industries, namely industries in which the national female employment share is above the cross-industry median. That is, we compute the following:

$$GF_{a,t,t+j} = \frac{1}{j}\frac{1}{2}\sum_{i}^{I} s_{a,i,t}\left|\frac{1+g_{a,i,t,t+j}}{1+g_{a,t,t+j}} - 1\right| * \mathbb{1}\left\{fs_{it} > \overline{fs_t}\right\}, \tag{2}$$

where all the terms are computed as above, fs_{it} indicates the female share in industry i in year t, and (with a slight abuse of notation) $\overline{fs_t}$ indicates the cross-industry *median* female share in year t.

Gender Segregation We measure gender segregation using the Duncan-Duncan Segregation Index (Duncan and Duncan, 1955), defined as

$$DDI = \frac{1}{2}\sum_{i=1}^{N}\left|\frac{m_i}{M} - \frac{f_i}{F}\right| \tag{3}$$

where $\frac{g_i}{G}$ is the share of gender g that is employed in industry i. If men and women are distributed equally across industries, the index is equal to zero, and if industries are entirely gender segregated, the index is equal to one. The index can also be thought of as the share of employment that would need to switch industries to remove all segregation. Importantly, occupational segregation by gender is correlated with gender-conservative attitudes as measured by the Gender Progressivity Index from the General Social Survey (Kleven, 2022) (Figure 2), which supports our use of the DDI as a proxy for gender norms. However, differently from the Gender Progressivity Index, we can measure it at a very granular level (commuting zone), which allows us to exploit within-state variation, and thus control for common state-level confounders.

2.2 PSID Data for Micro Analysis

For the micro-level analysis, we use data from the Panel Study of Income Dynamics (PSID), covering 1997-2021 (biennial waves). In ongoing analysis (not yet included in this draft), we also use comparable data from the National Longitudinal Survey of Youth 1979 (NLSY79).

Gender Attitudes Gender attitudes are elicited for primary caregivers in the Child Development Supplement: this is arguably a selected sample (individuals with a child aged 12 or younger at least once between 1997 and 2021), so that should be kept in mind when interpreting the results. Primary caregivers (usually mothers) are asked to express their agreement with a series of statements about gender roles, such as *"It is much better for everyone if the man earns the main living and the woman takes care of the home and family."* We combine 12 such statements via Principal Component Analysis; the first component serves as a gender progressivity index. We assign mothers' progressivity scores to their male spouses/partners, relying on the well-documented assortative mating on attitudes (Luo and Klohnen, 2005). We drop the 2.5% of male primary caregivers. We classify men as "traditional" if their assigned progressivity falls in the bottom tercile of this progressivity distribution, and "progressive" otherwise.

Layoff We define a *layoff* event as a job separation where the reason reported is either that the company went out of business or that the individual was laid off or fired. We classify industries and

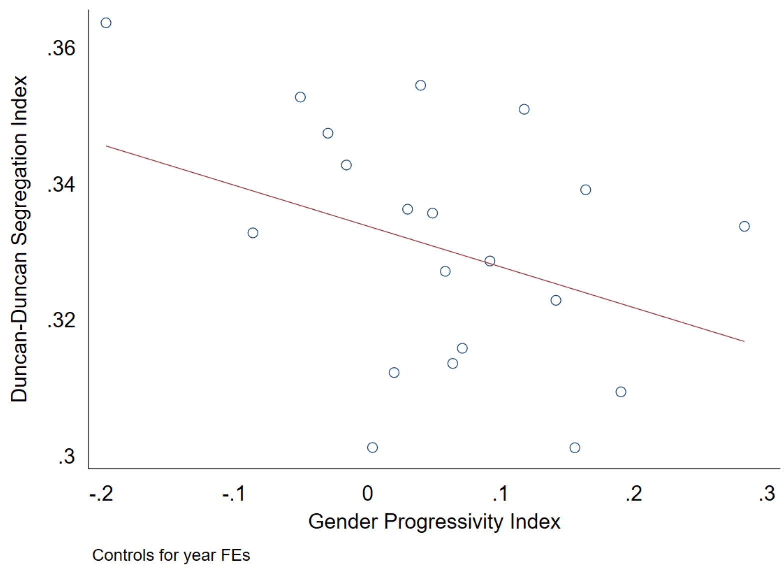


Figure 2: Correlation Between Gender Occupational Segregation and Gender Progressivity

Note: This figure shows the correlation (binscatter) between the occupational segregation by gender (Duncan-Duncan Index) and the Gender Progressivity Index across US states and decades (1990s-2010s). The Gender Progressivity Index summarizes the three gender norms-related questions that are consistently asked in the General Social Survey from the beginning (1972) to today and is constructed for states-by-decades by Kleven (2022). The segregation index is aggregated from commuting zone to state level by weighted average. The observations are at the state-by-decade level and are weighted by population. We control for year fixed effects.

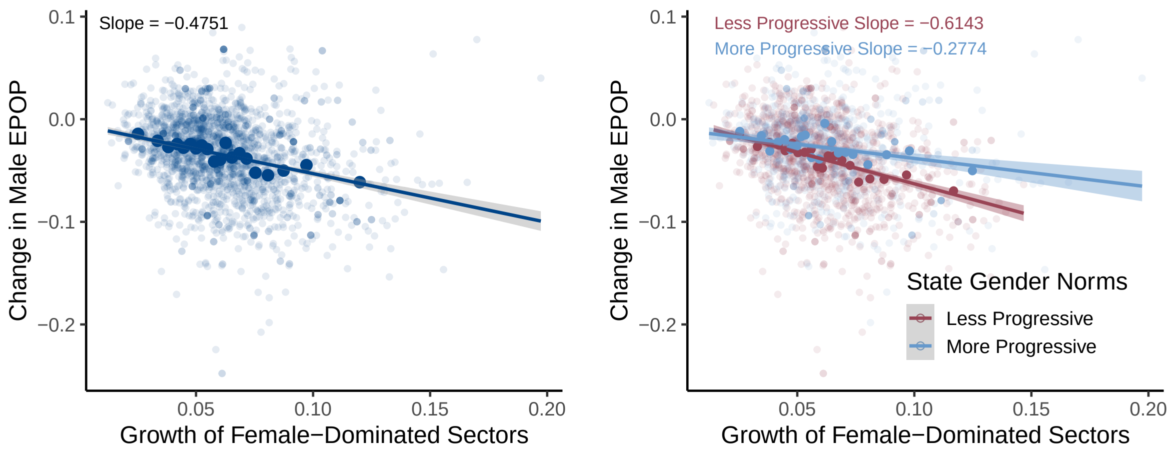
occupations as "female-dominated" if more than 49% (for industries) or 55% (for occupations) of their workers are female in the 2005 CPS, corresponding to the top tercile of female share.

Work-related Masculinity Norms In addition, the PSID asks primary caregivers in 1997 whether they agree with the statement: *"There is some work that is men's and some that is women's and they should not be doing each other's."* We classify mothers as "traditional" if they agree (13.7%) or strongly agree (4.6%) and as "progressive" if they disagree (52.2%) or strongly disagree (29.4%). We then examine whether children's occupational choices as adults (age 25) differ systematically by their mother's views.

2.3 Survey Data for Experimental Analysis

In our experimental analysis, we use Prolific to collect information on masculinity norms, exogenously vary perception on them, and elicit willingness-to-pay to avoid occupations perceived as feminine. While we are currently fielding this survey on Prolific in the US, and the pilot results refer to this sample, these questions will also be included as part of the Innovation Panel of the UK household longitudinal panel *Understanding Society* (University of Essex, Institute for Social and Economic Research, 2026).

As emphasized by Matavelli et al. (2026), while over time and across countries extensive data has been collected on gender norms prescribing women's behavior, very little information has been collected to date about norms prescribing men's behavior (with their paper as the notable exception). Therefore, in our survey, we elicit masculinity norms, alongside a vignette experiment to elicit willingness-to-pay to avoid masculinity norms. Section 5 provides more details.



(a) Aggregate

(b) By State Progressivity

Figure 3: Correlation between Male EPOP and Growth of Female-Dominated Sectors

Note: This figure shows the correlation (scatter, in lighter shades, and binscatter, larger and in darker shades) between the decadal change in Employment-to-Population ratio for men and the decadal growth of female-dominated sectors (as defined in Section 2.1), both in the aggregate (panel a) and separately by state progressivity at beginning of the decade (measured from the General Social Survey via Kleven, 2022; panel b). The observations are are the commuting zone-by-decade level. For the binscatter and the best-fit line, observations are weighted by population.

3 Macro Evidence: Norms, Segregation, and Reallocation

3.1 Suggestive Patterns

Figure 1 displayed a contemporaneous growth in female-dominated sectors and decline in male employment: here we show that the two trends are connected, and linked by gender norms. Panel (a) of Figure 3 shows the negative correlation between the decennial growth in female-dominated sectors and the decennial change in male employment-to-population ratio across commuting zones, between 1980 and 2010. The relationship is quite strong: a 10 percentage point excess-increase in growth in female-dominated sectors is correlated with a 4.7 percentage point decline in male employment-to-populatio ratio. As shown by panel (b), this relationship is twice as strong in less progressive states, where a 10 percentage point excess-increase in growth in female-dominated sectors is correlated with a 6.1 percentage point decline in male employment-to-population ratio, than in more progressive states, where it is correlated with a decline by 2.7 percentage points.

These correlations are suggestive, but could be driven by other underlying factors we are not accounting for: this is why we move to an IV-setup and more granular norms in the rest of this section.

3.2 Empirical Method

To study whether the effects of reallocation differ in more and less segregated labor markets, where greater segregation corresponds to lower progressivity, we estimate the effects of the interaction between reallocation and segregation on the employment-to-population ratio separately for men and women. To do so, we estimate the following regression:

$$\Delta EPop_{a,t,t+j} = \alpha + \alpha_1 R_{a,t,t+j} + \alpha_2 DDI_{a,t} + \beta R_{a,t,t+j} \times DDI_{a,t} + \alpha_3 gb_{t,t+j} + \varepsilon_{a,t,t+j}. \quad (4)$$

where $R_{a,t,t+j}$ is our measure of reallocation and $DDI_{a,t}$ is our measure of gender segregation by industry, both described in Section 2.1. For easier interpretation, and because we do not necessarily expect the effect of segregation to be linear, $DDI_{a,t}$ is here an indicator for commuting zones with above-median segregation. In this specification, α_1 is the effect of reallocation on the change in employment-to-population ratio in areas with below-median segregation, and β is the difference in this relationship in areas with above-median segregation. Put differently, our coefficient of interest, β, measures the effect of segregation on the effect of reallocation on the change in employment. We control for local area population density and educational composition, Census region, and Bartik-predicted employment growth. We weight the observations by log commuting zone population.

To account for the endogeneity of reallocation, we follow Chodorow-Reich and Wieland (2020) by using a Bartik instrument that uses local initial industry shares but replaces local growth rates with national growth rates:

$$Rb_{a,t,t+j} = \frac{1}{j}\frac{1}{2}\sum_{i}^{I} s_{a,i,t}\left|\frac{1+g_{i,t,t+j}}{1+g_{t,t+j}} - 1\right|, \tag{5}$$

where $g_{i,t,t+j}$ is the national growth rate of industry i.

Recall that over this period, reallocation tended to be *female-biased*: employment moved disproportionately into female-dominated sectors (Figure 1).

3.3 Results

Table 1 shows the results of estimating equation (4). The coefficient on reallocation is negative: reallocation (which, as we noted, was mostly female-biased over this period) is associated with a decline in male employment. More importantly for our story, the coefficient on the interaction between reallocation and segregation is negative, and almost as large as the coefficient on reallocation: reallocation decreases male employment more—almost twice as much—in areas that have above-median segregation. Instrumenting for reallocation with Bartik-predicted reallocation makes the effect less significant, but it remains negative and becomes larger in magnitude.

To put these numbers into context, consider two commuting zones, one above median segregation and one with below median segregation. If the low segregation commuting zone experienced reallocation across industries at the 75th percentile, we would expect the male employment-population ratio to decrease by 3.1 percentage points more than if it experienced reallocation at the 25th percentile. However, for the high segregation commuting zone, we would expect its male e-pop to decrease by 4.4 percentage points more in the high-reallocation scenario vs. the low-reallocation scenario. Thus, for a similar change in reallocation, the high-segregation area would see a 1.3 percentage point larger decrease in male employment. This is approximately equivalent to moving from the median of male employment changes to the 35th percentile.

Notably, the negative effects of segregation on participation seem to be present only for men, not women. Table 2 shows the same regressions, but with changes in the female employment-population ratio as the dependent variable. Although the baseline relationship between reallocation and participation is consistently negative for women, the interaction term is positive, but insignificant in the IV case.

There are several possible explanations. One, which we are particularly interested in, is that segregation has been particularly harmful for men because of both gender norms and the general trend that reallocation has been away from goods-producing and towards services-providing industries

Table 1: Change in Male Employment-Population Ratio vs. Reallocation x Segregation

	Dependent variable:			
	Δ Male EPOP$_{t,t+j}$			
	OLS		*instrumental variable*	
	(1)	(2)	(3)	(4)
Reallocation$_{t,t+j}$	-0.976^{***}	-0.895^{***}	-6.314^{***}	-5.284^{***}
	(0.236)	(0.241)	(2.080)	(1.555)
Growth$_{t,t+j}$ (Bartik pred.)	2.756^{***}	2.037^{***}	0.192	-0.799
	(0.277)	(0.329)	(0.859)	(0.833)
Segregation$_t$	0.010^{*}	0.008	0.039^{*}	0.035^{*}
	(0.005)	(0.005)	(0.021)	(0.020)
Reallocation$_{t,t+j}$ x Segregation$_t$	-0.740^{**}	-0.772^{**}	-2.324^{*}	-2.260^{*}
	(0.326)	(0.322)	(1.368)	(1.291)
Constant	-0.049^{***}	-0.039^{***}	0.082^{*}	0.070^{**}
	(0.007)	(0.007)	(0.046)	(0.034)
Controls	No	Yes	No	Yes
Observations	2,142	2,142	2,142	2,142
R^2	0.177	0.234	-0.343	-0.091
Note:	*p<0.1; **p<0.05; ***p<0.01			

This table shows regressions of the change in the male employment-population ratio on reallocation and segregation at the commuting zone level for the period 1980-2010, with observations every 10 years. The data are from the US Census via IPUMS. Columns 1 and 2 use OLS, and columns 3 and 4 use 2SLS where we use Bartik-predicted reallocation as an instrument for local reallocation. All columns include year fixed effects and are weighted by log commuting zone population. Columns 2 and 4 also include Census region fixed effects, an indicator for above-mean density (measured in 2000), and an indicator for above-median share with a bachelor's degree.

documented in Figure 1. Indeed, at the macro-level, occupational segregation by gender and progressive gender norms are negatively correlated (Figure 2), which might indicate gender-conservative norms as determinant of both the segregation at the steady-state (e.g. as suggested by Pan, 2015) and of the inability to adapt to a changing labor market. Importantly, the results are robust to controlling for the initial manufacturing share in the commuting zone, suggesting that the findings are not driven purely by the decline of manufacturing. We explore this question further at the micro level in the next section.

3.4 Heterogeneity by Age

We further examine whether the effects of reallocation and segregation vary by age group. Table 3 shows the IV results separately by age: the youngest men (aged 16-24), prime-age men (aged 25-44), and older men (aged 45-64). The baseline effect of reallocation on employment is largest for older men who have already accumulated industry-specific human capital, and is not particularly harmful for the youngest men, who have not yet committed to a specific sector. However, the interaction between reallocation and segregation tells a strikingly different story: it is largest in magnitude for the youngest

Table 2: Change in Female Employment-Population Ratio vs. Reallocation x Segregation

	Dependent variable:			
	Δ Female EPOP$_{t,t+j}$			
	OLS		*instrumental variable*	
	(1)	(2)	(3)	(4)
Reallocation$_{t,t+j}$	-0.790^{***}	-0.593^{***}	-7.244^{***}	-6.616^{***}
	(0.211)	(0.212)	(2.066)	(1.508)
Growth$_{t,t+j}$ (Bartik pred.)	1.717^{***}	2.080^{***}	-1.251	-1.238
	(0.247)	(0.288)	(0.853)	(0.808)
Segregation$_t$	0.003	-0.004	0.026	0.002
	(0.005)	(0.005)	(0.020)	(0.020)
Reallocation$_{t,t+j}$ x Segregation$_t$	1.259^{***}	1.407^{***}	0.160	1.321
	(0.291)	(0.282)	(1.358)	(1.252)
Constant	0.084^{***}	0.091^{***}	0.239^{***}	0.232^{***}
	(0.006)	(0.006)	(0.046)	(0.033)
Controls	No	Yes	No	Yes
Observations	2,142	2,142	2,142	2,142
R^2	0.740	0.767	0.475	0.593
Note:				*p<0.1; **p<0.05; ***p<0.01

This table shows regressions of the change in the female employment-population ratio on reallocation and segregation at the commuting zone level for the period 1980-2010, with observations every 10 years. The data are from the US Census via IPUMS. Columns 1 and 2 use OLS, and columns 3 and 4 use 2SLS where we use Bartik-predicted reallocation as an instrument for local reallocation. All columns include year fixed effects and are weighted by log commuting zone population. Columns 2 and 4 also include Census region fixed effects, an indicator for above-mean density (measured in 2000), and an indicator for above-median share with a bachelor's degree.

men and weakest for the oldest. This pattern is consistent with a story in which human capital and the difficulty of re-skilling dominate for older workers—reallocation is bad for them regardless of the level of segregation—while norms matter most for younger workers. Young men have less occupation-specific human capital and could in principle train for any growing sector; they do not face a re-skilling barrier, but they do face a norms barrier. If their local labor market is more gender-segregated (and hence more gender-traditional), they are less likely to enter the growing female-dominated sectors, even though they could.

Table 3: Change in Male Employment-Population Ratio by Age Group (IV)

	16-24	25-34	35-44	45-54	55+
	(1)	(2)	(3)	(4)	(5)
Reallocation$_{t,t+j}$	0.41**	−0.48***	−0.60***	−0.15	−0.57***
	(0.17)	(0.13)	(0.12)	(0.10)	(0.11)
Segregated$_t$	0.09***	0.02	−0.02	0.03**	−0.005
	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)
Reallocation$_{t,t+j}$ x Segregated$_t$	−0.66***	−0.13	0.14	−0.23**	0.02
	(0.17)	(0.13)	(0.12)	(0.10)	(0.11)
Observations	2,132	2,132	2,132	2,132	2,132
R^2	0.08	0.16	0.22	0.24	0.51

Note: *p<0.1; **p<0.05; ***p<0.01

This table shows IV regressions of the change in the male employment-population ratio on reallocation and segregation at the commuting zone level for the period 1980-2010, separately by age group. We use Bartik-predicted reallocation as an instrument for local reallocation. All columns include year fixed effects, Census region fixed effects, an indicator for above-mean density (measured in 2000), and an indicator for above-median share with a bachelor's degree. Weights: log commuting zone population.

4 Micro Evidence: Individual Norms and Labor Market Outcomes

The macro evidence presented above establishes that segregation—as a proxy for gender norms—amplifies the negative employment effects of industrial reallocation for men. However, the aggregate analysis cannot pin down whether this pattern is driven by gender norms specifically, as many other factors are correlated with segregation. In this section, we turn to individual-level data from the PSID to provide more direct evidence that gender norms shape men's labor market behavior.

4.1 Response to Layoff

We study whether the impact of a layoff differs between gender-traditional and gender-progressive men. Using the PSID, we estimate an event-study around layoff events using the Callaway and Sant'Anna (2020) estimator, which addresses concerns about heterogeneous treatment effects in staggered difference-in-differences designs. We include individual, year, and age fixed effects; the control group consists of men who are never laid off during our sample period. We estimate the model separately for gender-traditional and gender-progressive men, as defined by the bottom versus top two terciles of the assigned progressivity index. In this version, we don't add further controls; versions that include controls for pre-layoff occupation, industry, state, and year of layoff interacted with the post-layoff dummy yield the same, if not stronger, results (not yet included in this draft).

Figure 4 shows the event-study estimates for employment. Following a layoff, both traditional and progressive men experience a sharp decline in employment. However, employment recovers somewhat faster for progressive men. Figures 5 and 6 shed light on the mechanism: progressive men are more likely to switch into female-dominated industries and occupations after a layoff (despite the fact that they were more likely to be in such industries and occupations to start with). Traditional men, by

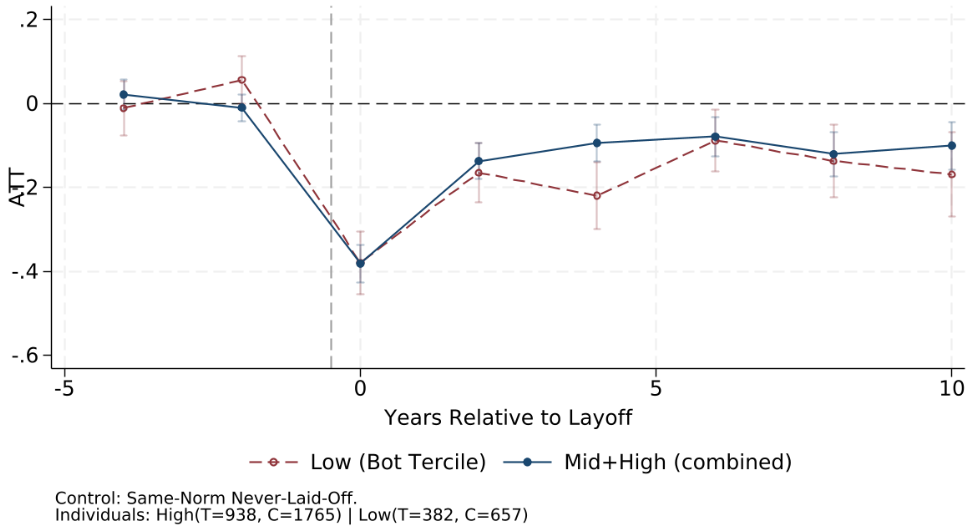


Figure 4: Employment Recovery After Layoff, by Gender Progressivity

Note: Event-study estimates of the effect of layoff on employment, estimated separately for gender-traditional (bottom tercile of progressivity) and gender-progressive (top two terciles) men. Estimator: Callaway and Sant'Anna (2020). Controls: individual, year, and age fixed effects. Control group: men never laid off. Data: PSID, 1997-2021.

contrast, show little increase in employment in female-dominated sectors, consistent with masculinity norms acting as a barrier to cross-gender occupational mobility.

These results should be interpreted with appropriate caveats. The sample is selected (men who at some point during 1997-2021 have a wife/cohabiting female partner and a young child), and the gender attitudes we use are elicited from wives/partners, not the men themselves. While assortative mating on attitudes is well-documented, the wife's attitudes are only a noisy proxy for the husband's own masculinity norms, which likely biases our estimates toward zero.

4.2 Intergenerational Transmission: Mother's Norms and Children's Occupation Choice

We next examine whether a mother's gender norms are correlated with gender segregation in occupational choice among her children. Using the direct question on "men's work and women's work" from the 1997 PSID, we classify mothers as traditional or progressive (as described in Section 2.2) and compute the Duncan-Duncan Index of segregation between their sons' and daughters' industries and occupations as adults (age 25, +/- 2 years). If a mother's norms have no bearing on her children's choices, the segregation between sons and daughters should be similar regardless of the mother's views.

Figures 7 and 8 show that sons and daughters of traditional mothers exhibit greater occupational and industry segregation than children of progressive mothers, consistent with the intergenerational transmission of gender norms shaping occupation choice. Interestingly, this pattern is driven by higher socioeconomic status (SES) families and by children of college-educated mothers: segregation between sons and daughters is significantly higher among children of traditional mothers only when the family is high-SES or the mother has a college degree. This is consistent with the idea that conforming to gender norms in occupational choice is a luxury: lower-SES families may face tighter economic constraints that override preferences for gender-typical work.

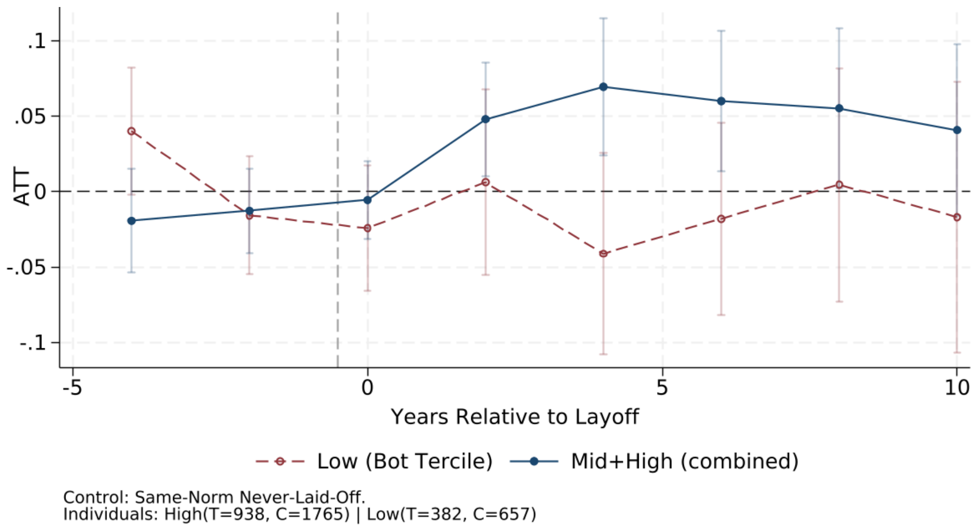


Figure 5: Switching to Female-Dominated Industries After Layoff

Note: Event-study estimates of the effect of layoff on the probability of working in a female-dominated industry (more than 49% female in 2005 CPS), estimated separately for traditional and progressive men. See notes to Figure 4 for details.

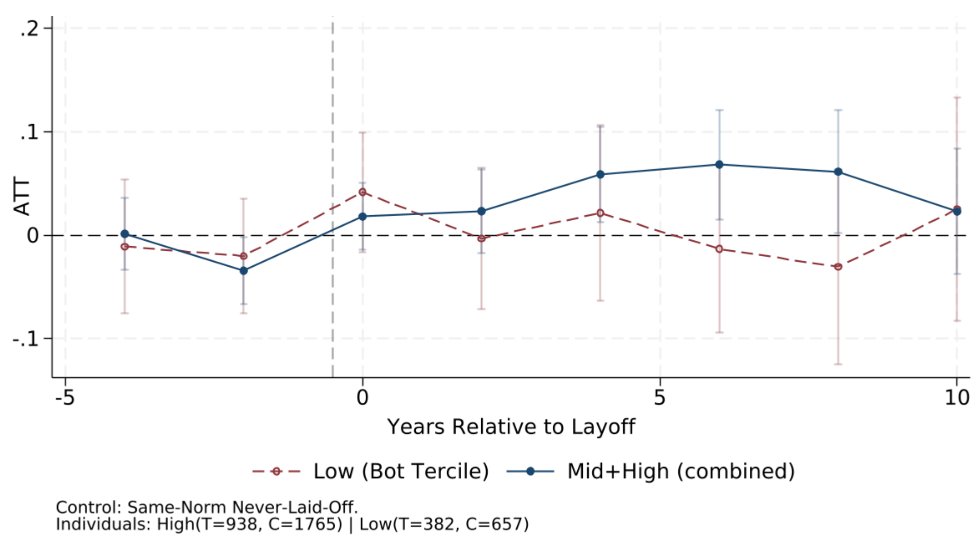


Figure 6: Switching to Female-Dominated Occupations After Layoff

Note: Event-study estimates of the effect of layoff on the probability of working in a female-dominated occupation (more than 55% female in 2005 CPS), estimated separately for traditional and progressive men. See notes to Figure 4 for details.

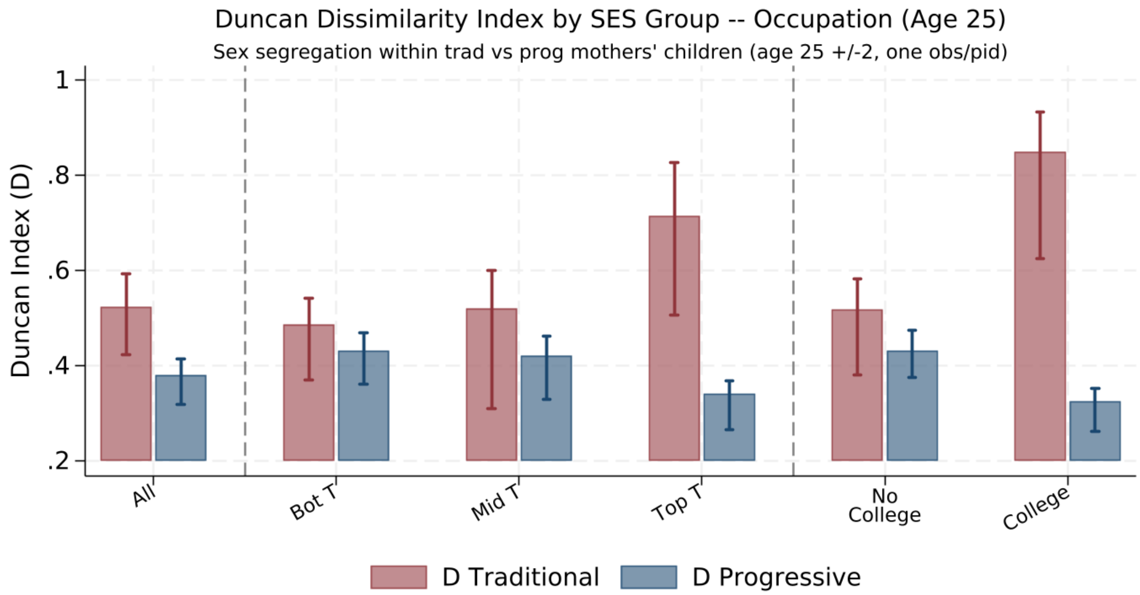


Figure 7: Occupational Segregation Between Sons and Daughters, by Mother's Norms

Note: Duncan-Duncan Index of gender segregation computed between sons' and daughters' occupations as adults (aged 25, +/- 2 years; one observation per individual), separately for children of "traditional" and "progressive" mothers. Mothers are classified based on their 1997 response to whether "there is some work that is men's and some that is women's." The leftmost panel aggregates children of all mothers respondents; the middle panel splits them by socioeconomic status (terciles of family income in 1997); the rightmost panel splits them by whether the mother has a college degree. Confidence intervals from bootstrap.

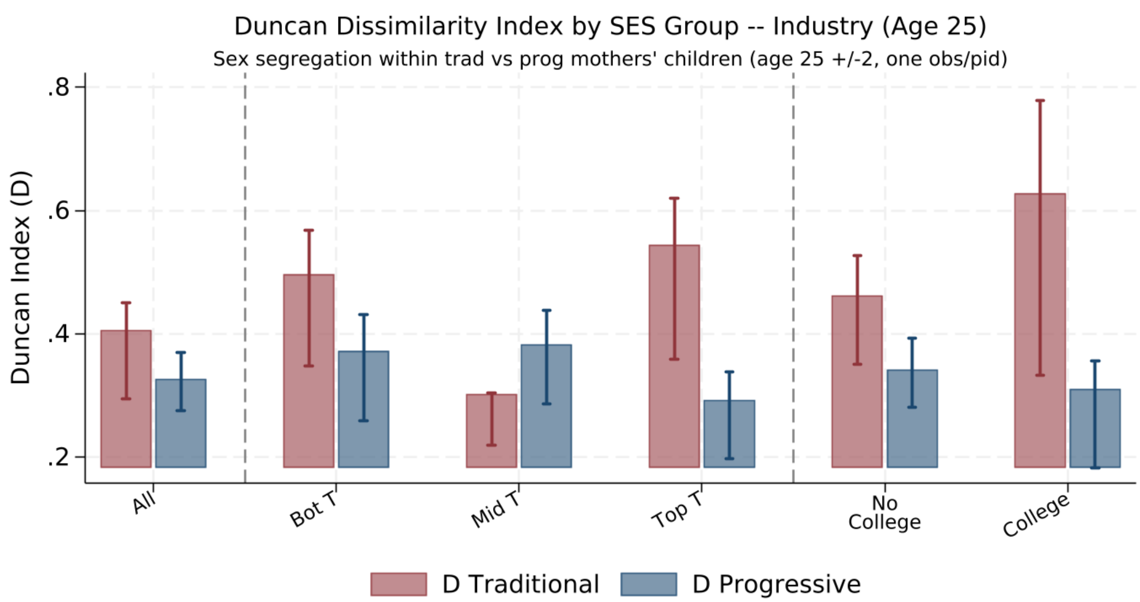


Figure 8: Occupational Segregation Between Sons and Daughters, by Mother's Norms

Note: Duncan-Duncan Index of gender segregation computed between sons' and daughters' industries as adults, separately for children of "traditional" and "progressive" mothers. See note to Figure 7 for details.

5 Experimental Evidence: Masculinity Norms and Occupation Choice

The macro and micro evidence presented above is suggestive but cannot establish a causal link between masculinity norms and occupational choice: we do not know whether *individual masculinity norms* specifically matter for occupation choice, how much (in dollars) they matter, or whether the relationship is causal. To address these questions, we are fielding a survey experiment designed to measure the relationship between masculinity norms and occupational preferences.

5.1 Survey Structure and Samples

We deploy the survey on two platforms: a US sample recruited through Prolific (US adults) and a UK sample embedded in the Understanding Society Innovation Panel (University of Essex, Institute for Social and Economic Research, 2026), the innovation module of the UK's main longitudinal household survey (analogous to the PSID). The UK version features occupation sets adapted to the British labor market. The survey consists of four modules: (0) demographics (education, employment, family structure), (1) masculinity norms elicitation and information treatment, (2) pairwise occupation choices for oneself, and (3) occupation ranking for a child or relevant young person.

5.2 Measuring Masculinity Norms

To measure masculinity norms, we ask respondents to provide their level of agreement with five statements regarding ideal masculine behavior following De Haas et al. (2024): (1) *Men should figure out personal problems on their own*; (2) *Men should be aggressive and competitive to get ahead*; (3) *It is important for a man to take risks*; (4) *Men should use violence to get respect if necessary*; (5) *Men should act strong even if scared inside*. We add a sixth statement directly related to occupation choice: *"Men should not work in occupations that are traditionally seen as feminine."*

5.3 Belief Elicitation and Information Treatment

Next, we ask respondents what fraction of 100 random adults (matching gender and education level of the respondent) would agree with one of the statements (drawn at random, oversampling the occupation norm statement). This elicits beliefs about the prevalence of masculinity norms in the population. We then randomly assign 50% of respondents to an information treatment: they are told the true fraction of respondents with their gender and education level who agreed or strongly agreed with the statement (similarly to Bursztyn et al., 2020). The treatment is inherently heterogeneous, as its direction depends on whether the respondent initially over- or under-estimated the prevalence of the norm. The remaining 50% (control group) receive no information.

To disentangle the effect of the information itself from the mere salience of thinking about masculinity norms before making occupational choices, half of the control group sees the masculinity norms elicitation module *after* (rather than before) the occupation choice and ranking modules. This allows us to test whether simply being prompted to reflect on masculinity norms—even without receiving any new information—affects subsequent occupational preferences.

5.4 Pairwise Occupation Choices

In the second part, respondents make a series of choices between pairs of hypothetical job offers. Jobs are drawn from traditionally masculine and feminine occupations, following Hancock et al. (2020), plus a gender-neutral occupation. We offer different occupation sets depending on educational attainment:

	No College Degree	College Degree
Feminine	Nurse	Nurse
	Secretary	Elementary School Teacher
Masculine	Construction Worker	Construction Manager
	Vehicle Mechanic	Firefighter
Neutral	Post Office Clerk	Loan Officer

Table 4: Occupations in Pairwise Choice Module

Each respondent is offered free training and a guaranteed job in each occupation. We randomly vary the length of training (3, 6, or 9 months for non-college; 6, 9, or 12 months for college) and the wage (intervals of 5% from the top wage in each set, which is 50,000$ a year for non-college and 70,000$ a year for college). Respondents make 6 pairwise job choices plus an attention check. From these choices, we can calculate the willingness-to-pay (WTP) among men (and women) to avoid a feminine (or masculine) job. Coupled with the questions and information treatment on masculinity norms, we can assess whether stronger masculine norms correlate with a higher WTP to work in a gender-conforming occupation and whether correcting misperceptions about societal masculinity norms changes this WTP.

5.5 Occupation Ranking for a Child

Finally, respondents are asked to express their preferences among future occupations for a relevant young person. We show respondents a table with four occupations that vary in pay, prestige, and gender composition—a low-pay neutral occupation (e.g. waiter), a feminine occupation (e.g. registered nurse), a masculine occupation (e.g. firefighter) matched in pay to the feminine one, and a high-pay neutral occupation (e.g. doctor)—along with typical wage ranges and projected employment growth. Respondents rank these occupations by how pleased they would be if the young person chose them.

The framing of the question adapts to the respondent's family situation: respondents with children under 21 are asked about their child; those with older children or no children are asked about a friend's child, a hypothetical future child, or themselves (if under 21). The wage ranges shown are randomly drawn from the overlap of the 40th and 60th percentiles of actual wages for those occupations, so that sometimes the masculine job has a higher wage and sometimes the feminine job does. In addition, we randomize whether respondents see projected employment growth from the Bureau of Labor Statistics alongside the wage information. This exercise allows us to measure whether gender norms affect choices for one's child differently than those for oneself, and also to abstract from differences in skills that are already established among working adults.

5.6 Pilot Results

In May 2026, we run a pilot version of the survey, sent to 207 US respondents on Prolific; the responses are encouraging.

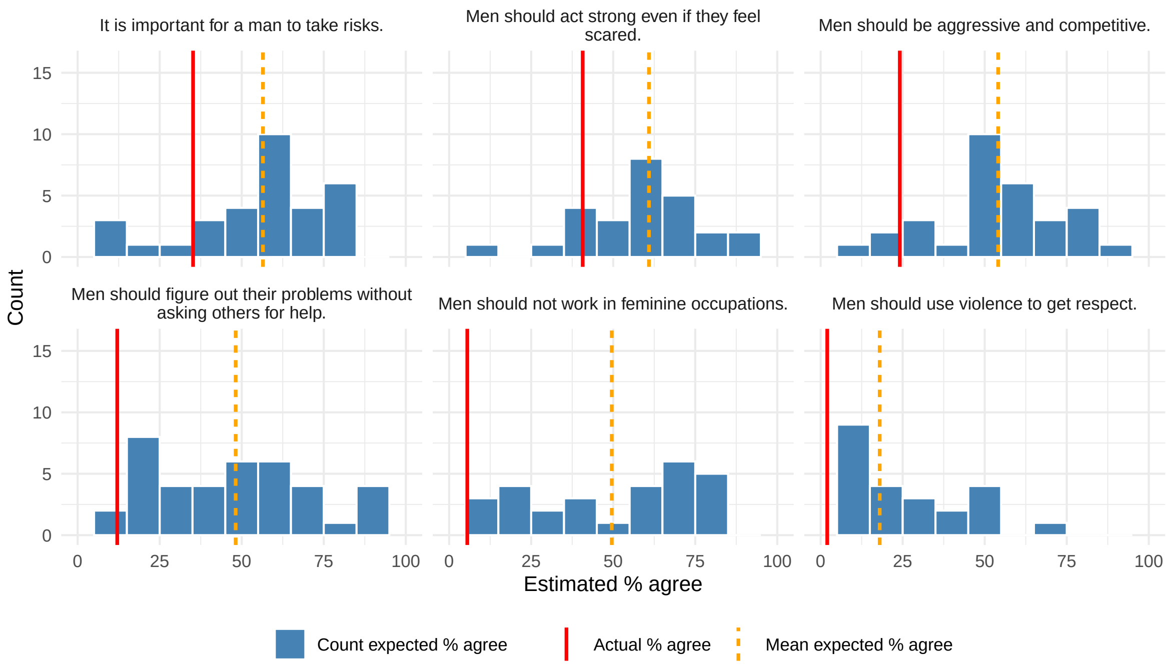


Figure 9: Actual Agreement with Masculinity Statements and Distributions of Beliefs

Note: For each of the six masculinity statements listed in Section 5.2, this figure displays the fraction of pilot respondents who agree or strongly agree with it (red solid vertical line), as well as the distribution of beliefs regarding the level of agreement in the relevant population (100 random Americans of the same gender and education level as the respondent) in blue. The yellow dashed vertical line is the average of the beliefs.

Demographics We explicitly over-sampled men (60%). The prolific sample tends to be highly educated (79%); 39% are married, 46% have kids, 78% are employed and the average age is 38. Out of the 207 respondents, 198 passed the attention check embedded in the elicitation of masculinity norms, and 181 the one embedded in the pairwise job comparison.

Measuring and Debiasing Masculinity Norms In the pilot sample, there is a varying degree of agreement with the masculinity norms statements and very dispersed beliefs about what others think, suggesting the presence of pluralistic ignorance and thus scope for a debiasing intervention (which we did not run in the pilot, as we did not have the "truth"). Figure 9 shows in red the fraction of respondents who agree with each statements, which varies from 41% for *"Men should act strong even if scared inside"* to 2% for *"Men should use violence to get respect"*. The same figure shows, in yellow, the average of the belief of how many people the respondent would expect to agree: for all statements, this is overestimated, and in certain cases by a margin of more than 40 percentage points. In addition, beliefs are very dispersed, with some respondents massively overestimating and some massively undestimating the true share of agreement. The pilot sample thus suggests the presence of pluralistic ignorance, and possible scope for a debiasing intervention.

Willingness-To-Pay to Avoid a Gender-nonconforming Occupation In the pilot sample, we observe a large willingness to pay by men to avoid being in a feminine job, which is very heterogeneous depending on revealed masculinity. Figure 10 shows that, on average, college-educated men are willing to pay 7,154$ a year (more than 10%) to avoid being in a female occupation relative to the gender neutral occupation (loan officer). However, this number decreases to 5,206$ for men who do not

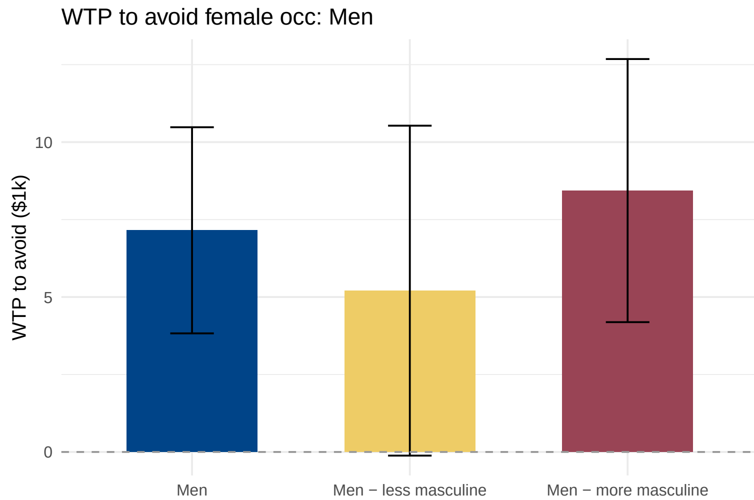


Figure 10: Willingness-to-pay to Avoid a Feminine Job Among Men in the Pilot Sample

Note: This figure shows the willingness-to-pay (WTP) to avoid a feminine job (elementary school teacher or nurse) relative to a gender neutral job (loan officer) among the sample of college-educated men in the pilot. The WTP is expressed in thousands of dollars per year. The WTP is computed as described in Section 5.4.

agree with any of the masculinity statements, while it is about 60% larger, 8,435$, for men who agree with at least one of the masculinity statements. Although in the pilot sample these differences are not statistically significant, the large difference suggests that men who adhere more strongly with masculinity norms do indeed perceive a large cost from going into female occupation, highlighting the presence of a friction that might be sizeable in the aggregate.

6 Conclusion and Next Steps

In this paper, we present evidence from multiple angles that masculinity norms act as a friction in occupational choice, contributing to the decline in male labor force participation. At the macro level, we show that sectoral reallocation leads to sharper declines in male employment in more gender-segregated areas, with the interaction between segregation and reallocation being particularly pronounced for the youngest men—precisely those who face the fewest human capital barriers to switching sectors. At the micro level, using individual-level data from the PSID, we find suggestive evidence that gender-progressive men's employment recovers faster after layoff, partly through switching into female-dominated industries and occupations, and that children of traditional mothers exhibit greater occupational segregation by gender—but only in higher-SES families that can "afford" to conform to norms.

These findings point to masculinity norms as a potentially important and under-explored barrier to efficient labor market adjustment. If masculinity norms are a primary factor in occupational choice, then providing men with training will not be sufficient to turn technicians displaced by the decline of manufacturing into well-paid nurses, and potentially prosperous reallocation will require other types of interventions.

Our ongoing survey experiment, currently in the pilot stage, will allow us to provide direct causal evidence on this question and to quantify the willingness to pay to avoid gender-atypical occupations. Through this survey coupled with our macro and micro evidence, we will be better able to assess whether and how much gender norms drive differing choices in the face of shifting demand.

References

Abraham, K. G. and M. S. Kearney (2020). Explaining the decline in the us employment-to-population ratio: A review of the evidence. Journal of Economic Literature 58(3), 585–643.

Aguiar, M., M. Bils, K. K. Charles, and E. Hurst (2021). Leisure luxuries and the labor supply of young men. Journal of Political Economy 129(2), 337–382.

Autor, D., D. Dorn, and G. Hanson (2019). When work disappears: Manufacturing decline and the falling marriage market value of young men. American Economic Review: Insights 1(2), 161–178.

Baranov, V., R. De Haas, and P. Grosjean (2023, September). Men. Male-biased sex ratios and masculinity norms: evidence from Australia's colonial past. Journal of Economic Growth 28(3), 339–396.

Binder, A. J. and J. Bound (2019, May). The declining labor market prospects of less-educated men. Journal of Economic Perspectives 33(2), 163–90.

Bursztyn, L., A. L. González, and D. Yanagizawa-Drott (2020, October). Misperceived Social Norms: Women Working Outside the Home in Saudi Arabia. American Economic Review 110(10), 2997–3029.

Callaway, B. and P. H. C. Sant'Anna (2020, December). Difference-in-Differences with multiple time periods. Journal of Econometrics.

Chodorow-Reich, G. and J. Wieland (2020). Secular labor reallocation and business cycles. Journal of Political Economy 128(6), 2245–2287.

Cortes, G. M., N. Jaimovich, and H. E. Siu (2023). The growing importance of social tasks in high-paying occupations: Implications for sorting. Journal of Human Resources 58(5), 1429–1451.

Cortés, P., J. Hwang, J. Pan, and U. Schönberg (2026, January). Gender norms and the labor market. Working Paper 34716, National Bureau of Economic Research.

De Haas, R., V. Baranov, I. Matavelli, and P. Grosjean (2024). Masculinity around the world. Centre for Economic Policy Research CEPR Discussion Paper DP19493.

Delfino, A. (2024, June). Breaking gender barriers: Experimental evidence on men in pink-collar jobs. American Economic Review 114(6), 1816–53.

Duncan, O. D. and B. Duncan (1955). A methodological analysis of segregation indexes. American Sociological Review 20(2), 210–217.

Hancock, A. J., H. M. Clarke, and K. A. Arnold (2020). Sexual orientation occupational stereotypes. Journal of Vocational Behavior 119, 103427.

Kleven, H. (2022). The geography of child penalties and gender norms: A pseudo-event study approach. Working Paper 30176, National Bureau of Economic Research.

Lilien, D. M. (1982). Sectoral shifts and cyclical unemployment. Journal of Political Economy 90(4), 777–793.

Luo, S. and E. C. Klohnen (2005). Assortative Mating and Marital Quality in Newlyweds: A Couple-Centered Approach. Journal of Personality and Social Psychology 88(2), 304–326.

Matavelli, I., P. Grosjean, R. D. Haas, and V. Baranov (2026, March). Masculinity Norms and Their Economic Implications.

Ngai, L. R. and B. Petrongolo (2017). Gender gaps and the rise of the service economy. American Economic Journal: Macroeconomics 9(4), 1–44.

Pan, J. (2015). Gender Segregation in Occupations: The Role of Tipping and Social Interactions. Journal of Labor Economics 33(2), 365–408. Publisher: The University of Chicago Press.

University of Essex, Institute for Social and Economic Research (2026). Understanding society: Innovation panel, waves 1-17, 2008-2024. DOI: http://doi.org/10.5255/UKDA-SN-6849-18. 15th Edition.